

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (5 points) Find the best fitting line $y = C + Dt$ for $b = 1, 2, 4$ at times $t = -1, 0, 2$.

(a) Write down the resulting equation $Ax = b$.

Solution:

$$\begin{array}{ccc} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} & \begin{bmatrix} C \\ D \end{bmatrix} & = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \\ \downarrow & \downarrow & \downarrow \\ A & x & b \end{array}$$

(b) Solve the normal equation for the least squares fitting, $A^T A \hat{x} = A^T b$, to find C and D .

Solution:

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{cases} C = 2 \\ D = 1 \end{cases}$$

Then the least squares line is

$$y = 2 + t$$

Problem 2. (5 points) Find an orthonormal basis q_1, q_2, q_3 for the column space of

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix} \text{ using the Gram-Schmidt process.}$$

Solution: $\tilde{q}_1 = a_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \|\tilde{q}_1\|^2 = 1$

$$\tilde{q}_2 = a_2 - \left(\frac{a_2^T \tilde{q}_1}{\tilde{q}_1^T \tilde{q}_1} \right) \tilde{q}_1$$

$$= \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \left(\frac{2}{1} \right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad a_2^T \tilde{q}_1 = 2$$

$$= \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \quad \|\tilde{q}_2\|^2 = 9$$

$$\tilde{q}_3 = a_3 - \left(\frac{a_3^T \tilde{q}_1}{\tilde{q}_1^T \tilde{q}_1} \right) \tilde{q}_1 - \left(\frac{a_3^T \tilde{q}_2}{\tilde{q}_2^T \tilde{q}_2} \right) \tilde{q}_2$$

$$= \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - \left(\frac{4}{1} \right) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \left(\frac{18}{9} \right) \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} \quad a_3^T \tilde{q}_1 = 4 \quad a_3^T \tilde{q}_2 = 18$$

$$\|\tilde{q}_3\|^2 = 25$$

$$\Rightarrow q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$q_3 = \frac{\tilde{q}_3}{\|\tilde{q}_3\|} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$